

Smart Meter NIC Hardware Capability

Author: Himanshu Gajjar

Date: 12/09//2024

Version: 0.1

Document No.: RF-NIC-HW

Document Approver: Arsad

Document Classification: External

Distribution and approvals are maintained in:

It is the responsibility of the user of this document to verify that it is the most current edition.

Preface

Privacy Information

This document may contain information of a sensitive nature. This information should not be given to any persons outside of the System Level Solutions (I) Pvt. Ltd. without prior consent.

Revision History

Name	Description of Changes	Date	Version
Naveen Sharma	First release	11/09/2024	1.0

Approval

Name	Position	Signature	Date
Himanshu Gajjar	Manager	email approval	Email date: 12/09/2024

Table of Index

1 Introduction.....4

2 Design & Feasibility.....4

3 Pre-developed Libraries & Firmware.....5

4 RF Testing Capability.....5

5 QC JIG Development and testing.....6

1 Introduction

This document represent the SLS hardware capabilities for the Smart Meter RF NIC card.

Right now, SLS is working with the 4 Nos Meter manufacture for RF NIC card.

We are using our own make Wisun Module based on Silicon Lab chipset EFR32FG28 which is WPC certified.

SLS has designed and maintains stock of RF modules certified for popular technologies, including LoRaWAN, Wi-SUN, Zigbee, Wi-Fi, and Bluetooth. Depending on the project requirements, we can select any of these modules to ensure optimal performance.

2 Design & Feasibility

Based on the customer's specifications, such as the form factor of meter constructions and technical drawings, the SLS Product Design Team performs a comprehensive feasibility analysis.

Following the feasibility review, NIC Design is initiated to meet the exact customer specifications and ensure smooth integration with their systems.

SLS has prepared one checklist based on experience with the different Meter manufacturer for RF NIC card development. Smart meter manufacturer need to provide those details to start the board design. Using this checklist, we can minimized the overall design timeline.

Below image shows the block diagram and we have readily design block's component available and base on the Meter manufacturer requirement, we design the circuit diagram and base on the mechanical requirement do the placement, select appropriate PCB antenna pattern and make the layout design accordingly.

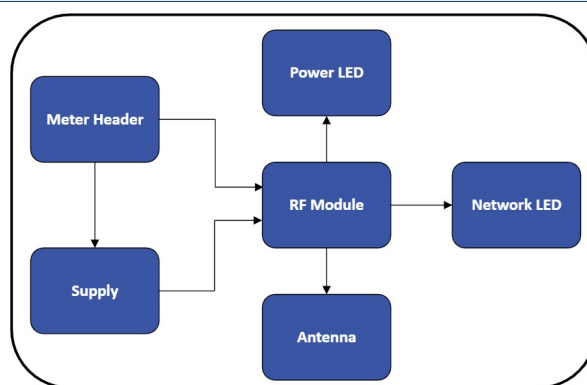


Figure 1: Block Diagram

3 Pre-developed Libraries & Firmware

SLS has developed and finalized libraries and firmware that are ready to integrate with our certified RF modules. This enables in-house testing and rapid deployment.

4 RF Testing Capability

We have our own subsidiary lab known as Avartanam Test Labs.

- RF Custom Antenna Design: Once the RF NIC feasibility is complete, SLS collaborates with Avartanam Test Labs for customized antenna design services tailored to specific needs.
- TRP & EIRP Measurement & Optimization: After the RF NIC board assembly, Avartanam Test Labs provides support with Total Radiated Power (TRP) and Effective Isotropic Radiated Power (EIRP) measurements. Additionally, they assist in optimizing these parameters using our in-house developed firmware.

By combining SLS's design capabilities with the testing and measurement expertise from Avartanam Test Labs, we ensure that all RF performance requirements are met efficiently and effectively.

Please refer our Avartanam lab PPT for more details.

5 QC JIG Development and testing.

SLS is a turnkey Electronics Manufacturing Service provider helping businesses with PCB assembly, prototyping, and small and medium volume production. We are equipped with semi-automated and fully-automated SMT lines, services for automated QC, and turnkey project handling.

To provide world-class electronics manufacturing services, we have developed a state-of-the-art Test Automation product enabling Board Function Tests (BFTs) on assembled PCB's and Quality Check (QC) on final products. This enables us in helping our clients with quality tested, standard products and help them create a niche for their products in the market.

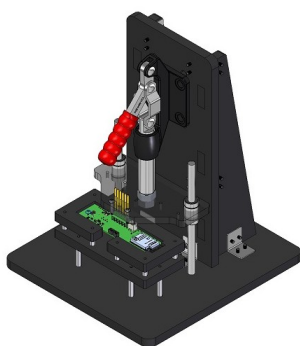


Figure 3: Jig Image1

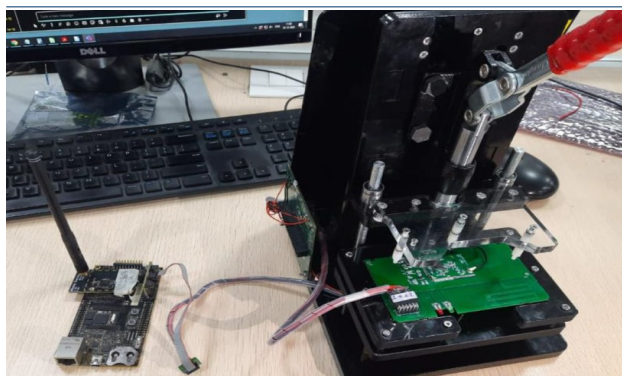


Figure 2: Jig image2

Indicator	Test Class	Test Name	Status	Result	Test Duration(m:ss.FF)
✓	10005	Get board unit variant	Completed	Passed	00:00:21
✓	10008	Get Mac Address of device	Completed	Passed	00:00:18
✓	10007	Get hardware version (PCB Revision)	Completed	Passed	00:00:15
✓	11700	Validate ADC	Completed	Passed	00:00:22
✓	11209	Configure QC Controller GPIO with DUT GPIO	Completed	Passed	00:00:21
✓	11200_1	Perform All DUT Input GPIO Test	Completed	Passed	00:01:23
✓	11200	Perform All DUT Output GPIO Test	Completed	Passed	00:01:11
✓	10702_1	Set 2.4GHz channel for reference device	Completed	Passed	00:00:14
✓	10703_1	Set 2.4GHz power For reference device	Completed	Passed	00:00:12
✓	10704_1	Get 2.4GHz Config for reference device	Completed	Passed	00:00:22
✓	10702	Set 2.4GHz channel For DUT	Completed	Passed	00:00:20
✓	10703	Set 2.4GHz power For DUT	Completed	Passed	00:00:22
✓	10704	Get 2.4GHz Config for DUT	Completed	Passed	00:00:30
✓	10715	Validate Receiving - Conductive Functionality	Completed	Passed	00:03:26
✓	10715_1	Validate Receiving - Radiative Functionality	Completed	Passed	00:03:29
✓	10716	Frequency Offset Tuning Measurement	Completed	Passed	00:20:09
✓	10717	Get Frequency Offset Value	Completed	Passed	00:00:21

DUT Device Info: NM0112-P
Reference Device Mac: 84:2E:14:FF:FE:85:91:9A

HOST MAC: 84:2E:14:FF:FE:85:9C:47
NCP MAC:

Test Stamp: 14:55:45.645+03:30
[JCOM27] C10715 842x14fffe85919a 1
Test Stamp: 14:55:45.785+03:30
[JCOM27] R10715 Avg(CI:253 PER:0 RSSI: -98)
Test Stamp: 14:55:45.945+03:30
[JCOM27] C10710 1 0F 1
Test Stamp: 14:55:46.207+03:30
[JCOM27] R10710 P
Test Stamp: 14:55:46.344+03:30
[JCOM27] C10717 00
Test Stamp: 14:55:46.246+03:30
[JCOM27] R10717 HHHH2a
Test Stamp: 14:55:46.407+03:30
[JCOM27] C10710 1 10 01
Test Stamp: 14:55:46.485+03:30
[JCOM27] R10710 P
Test Stamp: 14:55:46.714+03:30
[JCOM27] C10710 0 0 0
Test Stamp: 14:55:46.229+03:30
[JCOM27] R10710 P

Len:23 payload: 10 61 cc 44 34 12 47 8c 85 fe ff
Packet 7 is sent
Len:23 payload: 10 61 cc 44 34 12 47 8c 85 fe ff
Packet 8 is sent
Len:23 payload: 10 61 cc 44 34 12 47 8c 85 fe ff
Packet 9 is sent
Len:23 payload: 10 61 cc 44 34 12 47 8c 85 fe ff
Packet 10 is sent
Len:23 payload: 10 61 cc 44 34 12 47 8c 85 fe ff
C10715 842x14fffe85919a 1
C10710 1 0F 1
C10717 00
Frequency Offset: -1750(dec)
C10710 1 10 01
C10710 0 0 0

Figure 4: Jig Application Reference Image